

DS 5110: Introduction to Data Management and Processing

Fall 2024

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Project: Customer Feedback Analysis and Summarization

Objectives:

- **PySpark DataFrames:** Master loading, filtering, and manipulating large text datasets.
- **NLP Techniques:** Apply fundamental NLP tasks, including sentiment analysis, text cleaning, and summarization.
- **Real-World Application:** Gain insight into analyzing customer feedback—a practical and widely used NLP application in industry.

Team Structure

Teams of 2-3 students are recommended, and you may work with the same group as your final project team. Each member can focus on a specific aspect of the project, such as data cleaning, sentiment analysis, summarization, or presentation. This project is designed to be completed over two class sessions, fostering collaboration and practical application of NLP skills to generate actionable business insights.

Dataset

Please go the link below and download the dataset for this project
<https://www.kaggle.com/datasets/cynthiarempel/amazon-us-customer-reviews-dataset?resource=download>

Instructions

1. **Data Loading and Setup:**
 - Start by loading the dataset into a PySpark DataFrame.
 - Display the structure of the data (schema) to understand the columns available for analysis.
2. **Data Preprocessing:**
 - **Tokenization and Cleaning:** Split reviews into sentences, convert text to lowercase, and remove punctuation and stop words.
 - **Filtering:** Filter out reviews with low ratings (1-2) to focus on negative feedback and those with high ratings (4-5) for positive feedback.
3. **Text Analysis and Processing:**
 - **Sentiment Analysis:**
 - Use basic word matching or a pre-trained NLP model in PySpark (if accessible) to determine the sentiment of each review.
 - Add a sentiment column with values such as "positive", "neutral", or "negative".
 - **Key Phrase Extraction:**

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- Extract key phrases or recurring terms related to common issues or positive highlights. For example, words like "quality," "delivery," "price," or specific product features.
 - Store these key phrases in a new column.
4. **Summarization:**
- Create a summary of each sentiment category by selecting the most informative sentences based on word count or frequency of key phrases.
 - Provide a brief summary for both positive and negative reviews, highlighting the main topics customers mentioned.
5. **Aggregation and Insights:**
- Group data by product_category and sentiment to identify common issues per category (e.g., "electronics" may have complaints about battery life).
 - Calculate and display statistics, such as the percentage of positive and negative reviews for each product category.

Expected Outcome

By the end of this project, students should produce:

1. A summarized dataset with columns for key phrases, sentiment, and summarized feedback.
2. A brief report or presentation with insights into customer feedback trends and actionable recommendations.